

Curriculum Vitae

Jae Hak Son

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Summary

I am an evolutionary biologist who aims to understand how the non-coding and coding sequences influence genomic architectures and complexity, with expertise in genetics, molecular biology, and genomics. I started my PhD training in evolutionary genetics by studying the polygenic sex determination and the early stages of sex chromosome evolution. I am currently working on host-transposons intragenomic conflicts and mechanisms that drive diverse evolutionary trajectories of sex chromosomes.

Education

- 2019 University of Houston Ph.D. in Biology and Biochemistry
Supervisor: Assistant Professor Richard P. Meisel
- 2013 Seoul National University M.S. in Biological Science
Supervisor: Associate Professor Piotr G. Jablonski
- 2011 Hanyang University B.S. (Honor) in Molecular Life Science
Supervisor: Professor Young Gyu Chai

Research Experience

- 2021 - Current Rutgers University Postdoctoral Associate
Advisor: Dr. Christopher E. Ellison
- 2019 - 2021 Yale School of Public Health Postdoctoral Associate
Advisor: Dr. Serap Aksoy

Publications

Published

* correspondence; † equal contribution

11. Li X, Visser S, **Son JH**, Geuverink E, Kivanç EN, Wu Y, Schmeing S, Pippel M, Anvar SY, Schenke MA, Marec F, Robinson MD, Meisel RP, Wimmer E, Zande L, Bopp D, Beukeboom LW (2024). Divergent evolution of male-determining loci on different proto-Y chromosomes of the housefly. **Nature Communications**, 15(1): 5984
10. Saarman NP, **Son JH**, Zhao H, Cosme LV, Kong Y, Li M, Wang S, Weiss BL, Echodu R, Opiro R, Serap Aksoy S, and Caccone A (2023). Genomic evidence of sex chromosome aneuploidy and infection-associated genotypes in the tsetse fly *Glossina fuscipes*, the major vector of African trypanosomiasis in Uganda. **Infection, Genetics and Evolution**, 105501
9. **Son JH**†, Weiss BL†, Schneider DI, Dera KM, Gstöttenmayer F, Opiro R, Echodu R, Saarman NP, Attardo GM, Onyango M, et al. (2021) Infection with endosymbiotic *Spiroplasma* disrupts tsetse (*Glossina fuscipes fuscipes*) metabolic and reproductive homeostasis. **PLoS pathogens**, 17(9): e1009539.
8. Yang L, Weiss BL, Williams AE, Aksoy E, de Silva Orfano A, **Son JH**, Wu Y, Vigneron A, Karakus M, Aksoy S (2021) Paratransgenic manipulation of a tsetse microRNA alters the physiological homeostasis of the fly's midgut environment. **PLoS pathogens**, 17(6): e1009475.
7. Adhikari K, **Son JH**, Rensink AH, Jaweria J, Bopp D, Beukeboom LW, Meisel RP (2021). Temperature-dependent effects of house fly proto-Y chromosomes on gene expression could be responsible for fitness differences that maintain polygenic sex determination. **Molecular Ecology** 30(22), 5704-5720
6. **Son JH***, Meisel RP* (2021) Gene-level, but not chromosome-wide, divergence between a very young house fly proto-Y chromosome and its homologous proto-X chromosome. **Molecular Biology and Evolution**, 38(2): 606-618
5. **Son JH**, Kohlbrenner T, Heinze S, Beukeboom LW, Bopp D, Meisel RP (2019) Minimal effects of proto-Y chromosomes on house fly gene expression in spite of evidence that selection maintains stable polygenic sex determination. **Genetics**, 213: 313-327

4. Meisel, RP, Davey T, **Son JH**, Gerry AC, Shono T, Scott JG (2016) Is Multifactorial Sex Determination in the House Fly, *Musca domestica* (L.), Stable Over Time? **Journal of Heredity**, 107(7): 615-625
3. Yang E, **Son JH**, Lee SI, Jablonski PG, Kim HY (2016) Water striders adjust leg movement speed to optimize takeoff velocity for their morphology. **Nature Communications**, 7(1), 13698
2. Koh JS, Yang E, Jung GP, Jung SP, **Son JH**, Lee SI, Jablonski PG, Wood RJ, Kim HY, Cho KJ (2015) Jumping on water: Surface tension-dominated jumping of water striders and robotic insects. **Science**, 349(6247): 517-521
1. **Son JH**, Han CS, Lee SI, Jablonski PG (2014) Effect of sex, hunger, and relative body size on the use of ripple signals in the interactions among water striders *Gerris latiaabdominis*. **Journal of Asia-Pacific Entomology**, 17(4): 653-658

Preprint (review/revision)

Son JH, Lawlor MA, Virani M, Cao W, Levine MT, Ellison C. (2025). Convergence and conflict among telomere-specialized transposons across 60 million years of Drosophilid evolution. *bioRxiv* (2nd revision under review, *Genome Research*)

Son JH (2026). Escape and survival: transposon adaptations in the face of insect host silencing. (under review, *Current Opinion in Insect Science*)

In Preparation

Son JH, Luecke D, Bista B, Manat Y, Cao W, Beukeboom LW, Bopp D, Ellison CE, Saelao P, Meisel, RP (In prep) The house fly polygenic sex determination system provides insight into the early stages of sex chromosome differentiation

Son JH, Volski M, Cao W, Ellison CE. (In prep) Evolutionary dynamics of the nuclear export factor gene family across the *Drosophila* genus

Grant

2018 Sigma Xi Grants-in-Aid of Research

Awards and Fellowships

2019	Cullen Fellowship Travel Grant, University of Houston
2018	Departmental Student Travel Award, University of Houston
2017	Cullen Fellowship Travel Grant, University of Houston
2016	Departmental Student Travel Award, University of Houston
2016	Travel Stipend for the 2016 International Congress of Entomology (ICE Program Enhancement Funds)
2014-2019	Graduate Tuition Fellowship, University of Houston
2014-2016	Presidential Cullen and Stella Ehrhardt Fellowship, University of Houston
2011-2012	Research Assistant Scholarship, Seoul National University
2010	Merit-based Scholarship, Hanyang University
2009	Mentorship Scholarship, Hanyang University
2005	Merit-based Scholarship, Hanyang University
2005	Student Service Scholarship, Hanyang University

Instructional Experience

2011, 2012	Teaching assistant, Laboratory of Biology
2014 - 2019	Teaching assistant, Laboratory of Biology, Genetics, Human Anatomy

Community and Outreach Activities

Peer Reviews: Communication Biology, Genetics, Genomics, Molecular Biology and Evolution, Journal of the Kansas Entomological Society

Membership of Professional Societies

Genetics Society of America

Society of Molecular Biology and Evolution

Student Mentorship

2023-2024	Justin Seo	Dr. Christopher Ellison lab, Rutgers University
2022	Mikaela Benkert	New Jersey STEM Network, Montclair State University
2017-2019	Dalia Aldin	Dr. Richard Meisel lab, University of Houston

Public Outreach

2016 - 2018	Volunteering supervisor, National Math and Science Competition hosted by Korean-American Scientists and Engineers Association (KSEA), South Texas/Houston Chapter, Houston, TX
2025 - current	Outreach Director, Korean-American Scientists and Engineers Association (KSEA) New Jersey Chapter, NJ

Academic Activities

Invited Presentations

- 2026 The Plant and Animal Genomics Conference, San Diego, CA, USA
Jae Hak Son, David Luecke, Basanta Bista, Yesbol Manat, Weihuan Cao, Leo Beukeboom, Daniel Bopp, Christopher E Ellison, Perot Saelao, Richard P Meisel. The house fly polygenic sex determination system provides insight into the early stages of sex chromosome differentiation.
- 2023 Agriculture and Life Sciences, Seoul National University, Seoul, Korea
Jae Hak Son. Sex chromosome evolution in the house fly and diversification of telomeres in *Drosophila*.
- 2016 XXV International Congress of Entomology, Orlando, FL, USA
Jae Hak Son, Chang Seok Han, Sang-Im Lee, Piotr G. Jablonski. Effect of sex, hunger, and relative body size on the use of ripple signals in the interactions among water striders *Gerris latiaabdominis*.

International / National / Regional Conference Oral Presentations

- 2026 New York Population Genetics Meeting, New York, USA
Jae Hak Son, David Luecke, Basanta Bista, Yesbol Manat, Weihuan Cao, Leo Beukeboom, Daniel Bopp, Christopher E Ellison, Perot Saelao, Richard P Meisel. The house fly polygenic sex determination system provides insight into the early stages of sex chromosome differentiation.
- 2024 The Allied Genetics Conference, Metro Washington, DC, USA
Jae Hak Son, Mia T. Levine, Christopher E. Ellison. Evolutionary diversification and repeated gene capture by telomeric retrotransposons across the *Drosophila* genus.
- 2022 Department of Genetics, Human Genetics Institute of New Jersey, Rutgers University, NJ, USA
Jae Hak Son, Christopher E. Ellison. Evolutionary diversification and repeated gene capture by telomeric retrotransposons across the *Drosophila* genus.
- 2018 Southeast Texas Evolutionary Genetics and Genomics Symposium, Houston, TX, USA
Jae Hak Son, Tea Kohlbrenner, Svenia D. Heinze, Daniel Bopp, Richard P. Meisel. Minimal effects of neo-Y chromosomes on transcriptomes of house flies in spite of evidence that selection across ecological habitats maintains stable polygenic sex determination.
- 2018 Houston Regional Ecology and Evolution Student Symposium, Houston, TX, USA
Jae Hak Son, Richard P. Meisel. Minimal effects of neo-Y chromosomes on transcriptomes of house flies in spite of evidence that selection across ecological habitats maintains stable polygenic sex determination.
- 2017 9th Annual Graduate Student Symposium, Houston, TX, USA
Jae Hak Son, Richard P. Meisel. Sex-determining genes have a greater effect on sexually differential expressions than sex chromosomes in the house fly.
- 2017 Houston Regional Ecology and Evolution Student Symposium, Houston, TX, USA
Jae Hak Son, Richard P. Meisel. Sex-determining genes have a greater effect on sexually differential expressions than sex chromosomes in the house fly.

- 2016 8th Annual Graduate Student Symposium, Houston, TX, USA
Jae Hak Son, Richard P. Meisel. Interactions between the male-determining genotype and development temperature influence the alternative splicing of the sex-determining gene in the house fly.
- 2015 Houston Regional Ecology and Evolution Symposium, Houston, TX, USA.
Jae Hak Son, Richard P. Meisel. A female-determining allele increases over generations in the lab in a population carrying multiple male-determining factors in the house fly.
- 2013 Korean Society of Applied Entomology Conference, Muju, Korea.
Jae Hak Son, Sang-Im Lee, Piotr G. Jablonski. Comparison of anti-predator strategies among three species of water striders

International / National / Regional Conference Poster Presentations

- 2025 66th Annual Drosophila Research, San Diego, CA, USA
Jae Hak Son, Maria Volski, Cao W, Christopher E. Ellison. Evolutionary dynamics of the nuclear export factor gene family across the *Drosophila* genus.
- 2022 Transposable Elements, Cold Spring Harbor, NY, USA
Jae Hak Son, Mia T. Levine, Christopher E. Ellison. Evolutionary diversification and repeated gene capture by telomeric retrotransposons across the *Drosophila* genus.
- 2022 63th Annual Drosophila Research, San Diego, CA, USA. Online
Jae Hak Son, Christopher E. Ellison. Evolutionary diversification and repeated gene capture by telomeric retrotransposons across the *Drosophila* genus.
- 2020 The Allied Genetics Conference, Online
Jae Hak Son, Richard P. Meisel. A very young Y chromosome is cryptically differentiated from a homologous X chromosome at gene sequence and expression levels
- 2019 New England Association of Parasitologists, New Haven, CT, USA
Jae Hak Son, Daniela I. Schneider, Serap Aksoy. *Spiroplasma* in the tsetse fly (Uganda): distribution and potential functional effects on controlling trypanosome infections.
- 2019 60th Annual Drosophila Research, Dallas, TX, USA
Jae Hak Son, Tea Kohlbrener, Svenia D. Heinze, Leo W. Beukeboom, Daniel Bopp, Richard P. Meisel. Minimal effects of proto-Y chromosomes on gene expression and cryptic X-Y differentiations in house flies
- 2018 West Gulf Coast Regional Conference (Korean-American Bio-Medical Symposium), Houston, TX, USA
Jae Hak Son, Tea Kohlbrener, Svenia D. Heinze, Daniel Bopp, Richard P. Meisel. Minimal effects of neo-Y chromosomes on transcriptomes of house flies in spite of evidence that selection across ecological habitats maintains stable polygenic sex determination.
- 2018 Population, Evolutionary, Quantitative Genetics Conference, Madison, WI, USA
Jae Hak Son, Tea Kohlbrener, Svenia D. Heinze, Daniel Bopp, Richard P. Meisel. Minimal effects of neo-Y chromosomes on transcriptomes of house flies in spite of evidence that selection across ecological habitats maintains stable polygenic sex determination.
- 2017 West Gulf Coast Regional Conference (Korean-American Bio-Medical Symposium), Houston, TX, USA
Jae Hak Son, Tea Kohlbrener, Svenia D. Heinze, Daniel Bopp, Richard P. Meisel. Sex determining genes have a greater effect on gene expression than sex chromosomes in the house fly
- 2017 The annual meeting of the Society for Molecular Biology and Evolution 2017, Austin, TX, USA
Jae Hak Son, Tea Kohlbrener, Svenia D. Heinze, Daniel Bopp, Richard P. Meisel. Sex determining genes have a greater effect on gene expression than sex chromosomes in the house fly
- 2014 6th Annual Graduate Student Symposium, Houston, TX, USA
Jae Hak Son, Erin S. Kelleher. Experimental evolution of the repression system for *P-element*
- 2012 14th International Behavioral Ecology Congress, Lund, Sweden.
Jae Hak Son, Sang-Im Lee, Piotr G. Jablonski. Comparison of predator avoidance strategy in three species of Water striders: the effect of morphology on escaping from the attack of predators
- 2011 66th Annual Meeting of the Korean Association of Biological Sciences, Chuncheon, Korea.
Jae Hak Son, Chang Seok Han, Sang-Im Lee, Piotr G. Jablonski. Ripple Signaling in males and females of a water strider.

Experience for a visiting scholar or research assistant

2016 Visiting student, Institute of Molecular Life Science, University of Zurich

(Principal investigator: Dr. Daniel Bopp)

2010 Research Assistant, Lab of Behavioral Ecology & Evolution, Seoul National University

- Project: Soft morphing locomotion mechanism of water striders

2009 Research Assistant, Lab of Behavior & Ecology, Ewha Woman University

- Project: Acoustic communication in Birds, tit family.